

10.TRANSFORM TECHNIQUES AND SYSTEM REPRESENTATION

Experiment 10.1

```
clc;
```

```
clear;
```

```
clf;
```

```
//=====
```

```
// Pole-Zero Plot
```

```
//=====
```

```
// Transfer Function Coefficients
```

```
b = [1 -0.5];    // Numerator
```

```
a = [1 -1.2 0.32]; // Denominator
```

```
// Calculate Zeros and Poles
```

```
z = roots(b);
```

```
p = roots(a);
```

```
//-----
```

```
// Axes
```

```
//-----
```

```
re = -2:0.01:2;
```

```
im = -2:0.01:2;
```

```
plot(re, zeros(re));    // Real axis
plot(zeros(im), im);    // Imaginary axis
```

```
//-----
```

```
// Unit Circle
```

```
//-----
```

```
theta = 0:0.01:2*%pi;
```

```
unit_circle_x = cos(theta);
```

```
unit_circle_y = sin(theta);
```

```
plot(unit_circle_x, unit_circle_y);
```

```
//-----
```

```
// Plot Zeros
```

```
//-----
```

```
plot(real(z), imag(z), 'o');
```

```
//-----
```

```
// Plot Poles
```

```
//-----
```

```
plot(real(p), imag(p), 'x');
```

```
//-----
```

```
// Labels
```

```
//-----
```

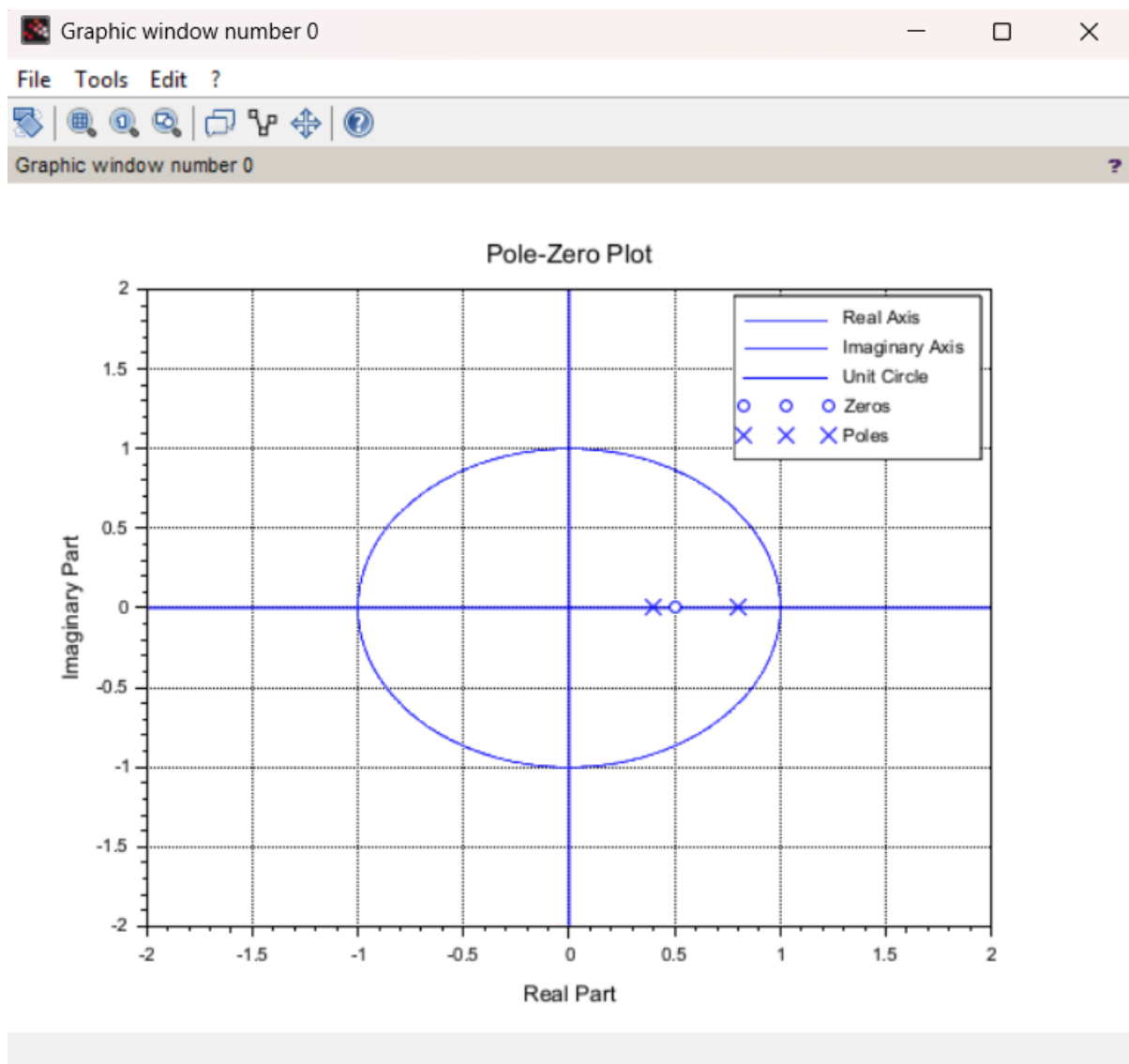
```
xlabel("Real Part");
```

```
ylabel("Imaginary Part");
```

```
title("Pole-Zero Plot");
```

```
legend("Real Axis","Imaginary Axis","Unit Circle","Zeros","Poles");
```

```
xgrid();
```



Experiment 10.2

```
clc;
```

```
clear;
```

```
clf;
```

```
//=====
```

```
// Numerical Z-Transform
```

```
//=====
```

```
// Input Sequence
```

```
x = [1 0.5 0.25 0.125];
```

```
// Number of Frequency Samples
```

```
N = 64;
```

```
// Time Index
```

```
n = 0:length(x)-1;
```

```
// Frequency Samples
```

```
k = 0:N-1;
```

```
omega = 2*%pi*k/N;
```

```
z = exp(%i*omega);
```

```
//-----
```

```

// Compute Z-Transform
//-----

Xz = zeros(1,N);

for m = 1:N
    Xz(m) = sum(x .* (z(m).^(-n)));
end

//-----
// Magnitude Spectrum
//-----
subplot(2,1,1);
plot(omega/(2*%pi), abs(Xz));
xlabel("Normalized Frequency");
ylabel("|X(z)|");
title("Magnitude Spectrum");

//-----
// Phase Spectrum
//-----
subplot(2,1,2);
plot(omega/(2*%pi), atan(imag(Xz), real(Xz)));
xlabel("Normalized Frequency");
ylabel("Phase (rad)");

```

```
title("Phase Spectrum");
```

```
//=====
```

```
// Numerical Inverse Z-Transform
```

```
//=====
```

```
// Recover Sequence using IFFT
```

```
x_rec = ifft(Xz);
```

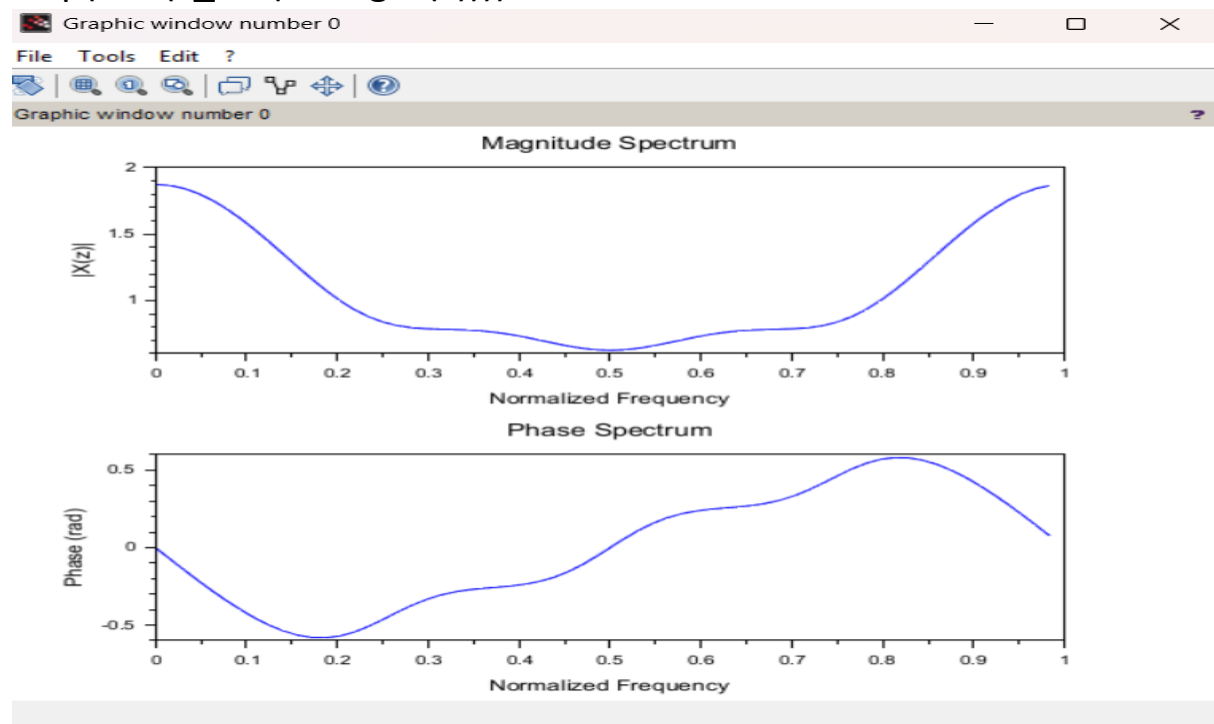
```
// Display Results
```

```
disp("Original Sequence:");
```

```
disp(x);
```

```
disp("Recovered Sequence:");
```

```
disp(real(x_rec(1:length(x))));
```



Experiment 10.3

```
// Test Case: Z-Transform of a Given Discrete-Time Signal
```

```
// Define the discrete time index
```

```
n = 0:3; // Time index from 0 to 3
```

```
// Define the discrete-time signal x[n]
```

```
x_n = [1, 2, 3, 4]; // Given signal
```

```
// Define the symbolic variable for Z
```

```
z = poly(0, 'z'); // Define a polynomial variable z
```

```
// Compute the Z-transform
```

```
X_z = 0; // Initialize Z-transform
```

```
for i = 1:length(n)
```

```
X_z = X_z + x_n(i) * z^(-n(i)); // Sum x[n] * z^(-n)
```

```
end
```

```

// Display the Z-transform expression
disp("Z-transform of x[n] = " + string(X_z));

// Define a range of z values on the unit circle
theta = 0:0.01:2*%pi; // Angle from 0 to 2*pi
z_vals = cos(theta) + %i * sin(theta); // Points on the unit circle

// Initialize arrays to store magnitude and phase
magnitude = zeros(1, length(theta));
phase = zeros(1, length(theta));

// Evaluate X(z) on the unit circle
for i = 1:length(z_vals)
    z_val = z_vals(i);
    X_z_val = 0;
    for j = 1:length(n)
        X_z_val = X_z_val + x_n(j) * z_val^(-n(j)); // Evaluate X(z) at z_vals(i)
    end
    magnitude(i) = abs(X_z_val); // Magnitude

    // Calculate phase using atan
    phase(i) = atan(imag(X_z_val) / real(X_z_val));
    if real(X_z_val) < 0 then
        phase(i) = phase(i) + %pi;
    end
end

// Plot magnitude and phase of the Z-transform
clf; // Clear the current figure

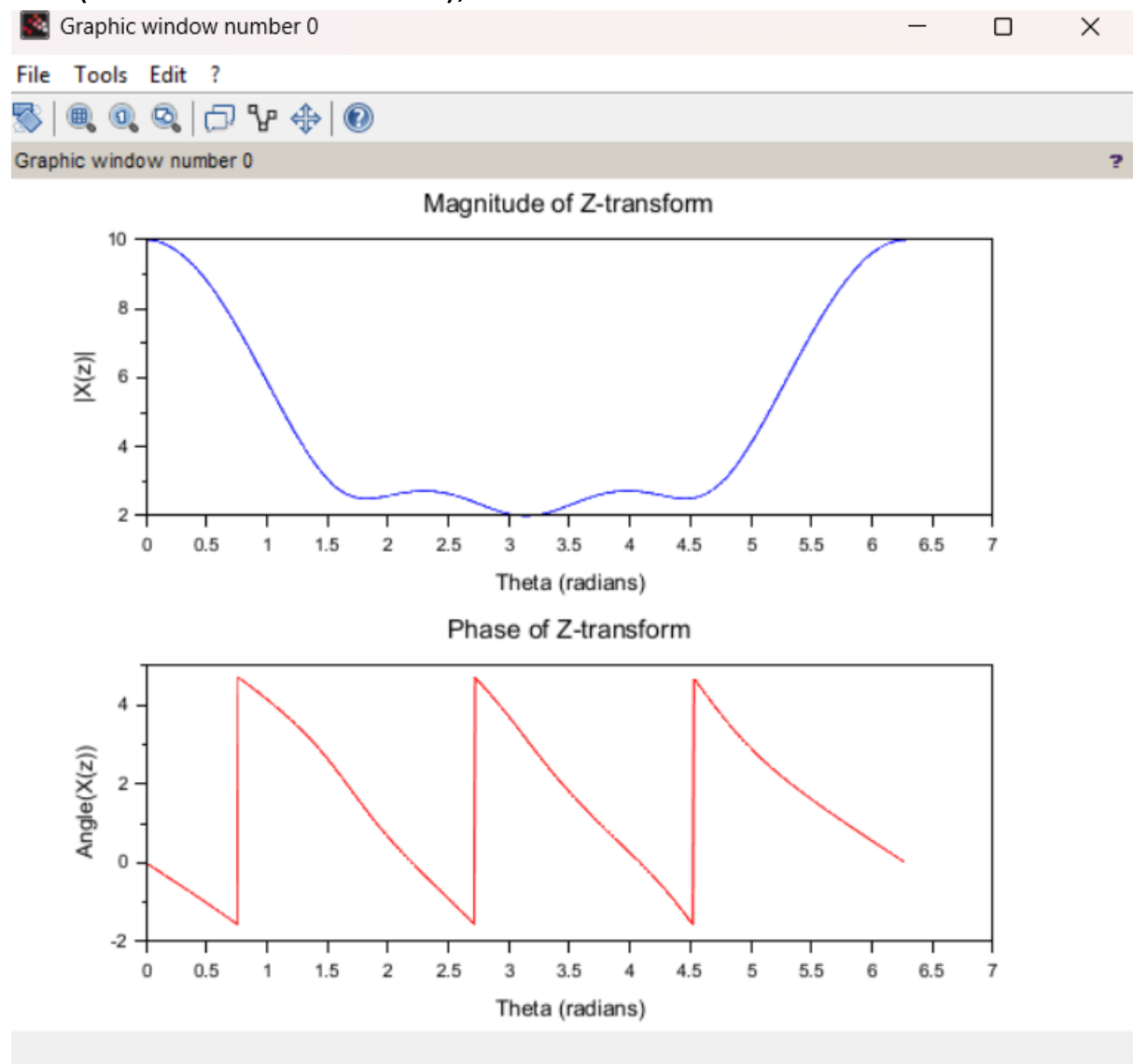
```



```
// Plot magnitude
subplot(2,1,1);
plot(theta, magnitude, 'b');
xlabel('Theta (radians)');
ylabel('|X(z)|');
title('Magnitude of Z-transform');
```

```
// Plot phase
subplot(2,1,2);
plot(theta, phase, 'r');
xlabel('Theta (radians)');
ylabel('Angle(X(z))');
```

```
title('Phase of Z-transform');
```



```
-> exec('C:\Users\teja\Documents\exp 1.sci', -1)
"Z-transform of x[n] = 4 +3z +2z^2 +z^3"
"Z-transform of x[n] = _____"
"Z-transform of x[n] =      z^3      "
```

Experiment 10.4

```
clc;
```

```
clear;
```

```
// Time vector (0 to 5 sec, step 0.001 s)
```

```
t = 0:0.001:5;
```

```

dt = t(2) - t(1); // time step
// Define continuous-time signal  $x(t) = e^{(-2t)} * \sin(3t)$ 
x = exp(-2*t) .* sin(3*t);
// s-domain values
sigma = 0; // Damping factor
omega = -20:0.5:20; // Frequency range
// Initialize Laplace magnitude array
X_mag = zeros(1, length(omega));
// Numerical Laplace calculation (manual trapezoidal
rule)
for k = 1:length(omega)
s = sigma + %i*omega(k);
Xs = x .* exp(-s*t);
// trapezoidal integration
X_mag(k) = abs( sum(Xs(1:$-1) + Xs(2:$)) * dt / 2 );
end
// Plot magnitude spectrum
clf;
plot(omega, X_mag, 'b');
xlabel("Frequency (rad/s)");

```

```
ylabel("|X(s)|");
```

```
title("Numerical Laplace Transform Magnitude");
```

